

Projector Phones

Great Way Of Sharing Personal Multimedia

Developments in projection technologies and optics have shrunk the size of the projector assembly to fit in mobile phones. Such projector phones, already demonstrated at *Consumer Electronics Show* and *Mobile World Congress*, are the next big thing to happen in the mobile phone market!

■ EFY BUREAU

As your cellphone allows you to capture high-resolution pictures and videos, there arises the need for a big screen to show them. Don't worry, with miniaturisation in projection technologies and optics, you can now carry in your pocket a screen as big as 127 cm (60 inches). The built-in projectors of cellphones allow you to project large images directly from your phone. This is a great way of sharing and enjoying digital content.

At last year's *Consumer Electronics Show* (CES), Texas Instruments (TI) had demonstrated digital light processing (DLP) based pico projector chips. This year's CES in January saw Samsung and LG integrating the DLP pico chip into their working mobile phones. Samsung's W7900 has an in-built DLP projector chipset, while LG's eXpo has an add-on pico-projector. Both the phones project images with resolution of 480×320 pixels.

Mobile World Congress (MWC) experienced more flavours of micro-projection technologies that can be built into

the cellphones. Samsung's Android operating system based Beam or I8520 Halo was one of the highlights. It uses TI's latest pico-projector chipset to give off an 800×600 projection with 10 lumens of brightness.

"Pico projectors come in handy while sharing photos and videos with a small group of people," says Ganesh S., business development manager, DLP products, Texas Instruments India. "Instead of sharing multimedia content on a 7.6cm (3-inch) phone display, one can project it as an A4-size image on any surface. Pico projection enhances the shared experience of viewing multimedia content from a handheld device," he adds.

Mobile projection technologies

Innovations in semiconductor technologies and optical electronics, and their integration in mobile devices to render large-screen experience, have started gaining attention from the manufacturers.

Projection requires a combination of several different technologies, ranging from scanning the image to translating it onto a big screen, so every player has something different to offer.

Microvision's PicoP uses a single micro-electromechanical system (MEMS) based light-scanning platform with coloured laser diodes for displaying the image. This system uses optical power as the intensity of each laser diode—red, green and blue—is adjusted



Optima projector phone